

Class 37 Darboux's Theorem

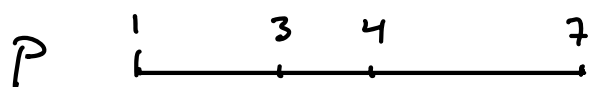
Goals

1. Know what it means for a function to be Riemann integrable on an interval $[a,b]$.
2. Use Darboux's Theorem to calculate the exact value of a definite integral.

Key idea: If f is integrable on the interval $[a,b]$ and P is any partition of $[a,b]$, then

$$L(f, P) \leq \int_a^b f dx \leq U(f, P)$$

Example: Find lower and upper bounds for the integral $\int_1^7 2-x dx$ given by the partition $P = \{1, 3, 4, 7\}$



interval	$M_i(f)$	$m_i(f)$	Δx_i	$M_i \Delta x_i$	$m_i \Delta x_i$
$[1, 3]$	1	-1	2	2	-2
$[3, 4]$	-1	-2	1	-1	-2
$[4, 7]$	-2	-5	3	-6	-15
				<u>-5</u>	<u>-19</u>
				$U(f, P)$	$L(f, P)$

$$\begin{aligned} \text{Actual value } \int_1^7 2-x dx &= 2x - \frac{1}{2}x^2 \Big|_1^7 = (14 - \frac{49}{2}) - (2 - \frac{1}{2}) \\ &= 12 - 24 = -12 \end{aligned}$$

If Q is a refinement of P , the lower and upper sums for Q form a tighter approximation for the integral.

Now let us consider all possible partitions of the interval $[a, b]$.

We denote the supremum of all the lower sums by $L(f)$ and the infimum of all the upper sums by $U(f)$.

Definition: A bounded function f defined on interval $[a, b]$ is Riemann integrable if $L(f) = U(f)$.

and we define $\int_a^b f(x) dx = L(f) = U(f)$

But it is too hard to calculate $U(f)$ and $L(f)$ by considering all partitions. We need a theorem to find the values a different way.

Darboux's Theorem: If f is any bounded function that is Riemann integrable on $[a, b]$ and if $\{P_n : n \in \mathbb{N}\}$ is a sequence of partitions of $[a, b]$ such that every subinterval of P_n has width less than or equal to $\frac{1}{n}$, then $U(f)$ and $L(f)$ both satisfy

$$U(f) = \inf \{U(f, P_n) : n \in \mathbb{N}\} = \lim_{n \rightarrow \infty} U(f, P_n)$$

and

$$L(f) = \sup \{L(f, P_n) : n \in \mathbb{N}\} = \lim_{n \rightarrow \infty} L(f, P_n)$$

Example: Use Darboux's Theorem to evaluate $\int_1^6 x + 1 dx$

Partition the interval $[1,6]$ into subintervals of width $\frac{1}{n}$

$$P_n = \left\{ 1, 1+\frac{1}{n}, 1+\frac{2}{n}, \dots, 1+\frac{5n}{n}=6 \right\}$$

upper height = $M_i = f\left(1+\frac{i}{n}\right) = 1+\frac{i}{n} = 2+\frac{i}{n}$

lower height = $m_i = f\left(1+\frac{i-1}{n}\right) = 1+\frac{i-1}{n} = 2+\frac{i-1}{n}$

width of each rectangle = $\frac{1}{n}$

area of upper rectangle = $\left(2+\frac{i}{n}\right) \cdot \frac{1}{n} = \frac{2}{n} + \frac{1}{n^2} i$

area of lower rectangle = $\left(2+\frac{i-1}{n}\right) \cdot \frac{1}{n} = \frac{2}{n} - \frac{1}{n^2} + \frac{1}{n^2} i$

$$U(f, P_n) = \sum_{i=1}^{5n} \left(\frac{2}{n} + \frac{1}{n^2} i \right) = \sum_{i=1}^{5n} \frac{2}{n} + \frac{1}{n^2} \sum_{i=1}^{5n} i$$

$$= 5n \cdot \frac{2}{n} + \frac{1}{n^2} \frac{5n(5n+1)}{2} = 10 + \frac{1}{2} \left(\frac{25n^2+5n}{n^2} \right)$$

$$= 10 + \frac{25}{2} + \frac{5}{2n} = 22.5 + \frac{5}{2n}$$

$$L(f, P_n) = \sum_{i=1}^{5n} \left(\frac{2}{n} - \frac{1}{n^2} + \frac{1}{n^2} i \right) = 5n \cdot \frac{2}{n} - 5n \frac{1}{n^2} + \frac{1}{n^2} \left(\frac{5n(5n+1)}{2} \right)$$

$$= 10 - \frac{5}{n} + \frac{25}{2} + \frac{5}{2n} = 22.5 - \frac{5}{2n}$$

$$U(f) = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} 22.5 + \frac{5}{2n} = 22.5$$

$$L(f) = \lim_{n \rightarrow \infty} L(f, P_n) = \lim_{n \rightarrow \infty} 22.5 - \frac{5}{2n} = 22.5$$

$$\int_1^6 x+1 dx$$

The Fundamental Theorem of Calculus

a) If f is continuous on $[a, b]$, then for every x in (a, b)

$$b) \quad \text{If} \quad \frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

$\int_a^b f(x) dx = F(b) - F(a)$
b) If f is continuous on $[a, b]$, and $F(x)$ is any antiderivative of $f(x)$, then

Part a) says that the rate of change of the accumulating function $\int_a^x f(t) dt$ is given by the value of the function f at any x in the domain of integration

Part b) says you can evaluate a definite integral of a continuous function by finding the difference in the values of an antiderivative at the boundaries of the domain of integration.

Use the definition of derivative to prove part a.
Then use part a to prove part b.